

EC-350 AI and Decision Support Systems

Week 5 Solving Problems by Searching

Dr. Arslan Shaukat



Acknowledgement: Lecture slides material from
Stuart Russell

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Iterative Deepening DFS

- General search strategy used with DFS
- We look for the solution at first level and then gradually increase the depth limit
- First 0, then 1, then 2 and so on
- Combines the benefits of BFS and DFS

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Iterative Deepening Search $l = 0$

Limit = 0



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Iterative Deepening Search $l = 1$

Limit = 1



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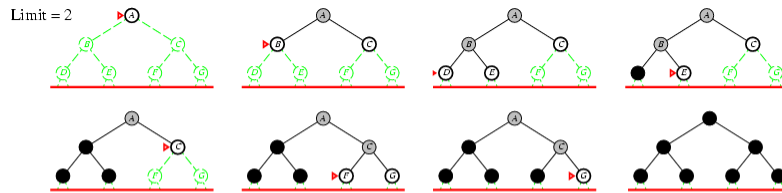
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Iterative Deepening Search $l = 2$



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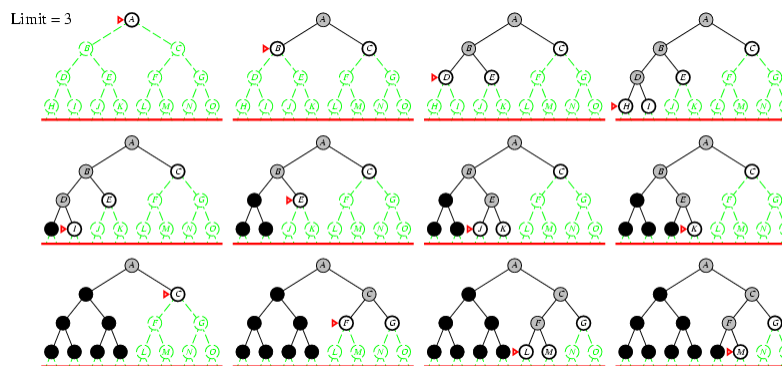
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Iterative Deepening Search $l = 3$



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Properties of Iterative Deepening Search

- Complete? Yes
- Time? $(d+1)b^0 + d b^1 + (d-1)b^2 + \dots + b^d = O(b^d)$
- Space? $O(bd)$
- Optimal? Yes, if step cost = 1

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Iterative Deepening Search

- No of nodes generated in a Breadth-first search to depth d with branching factor b :

$$N_{BFS} = b^0 + b^1 + b^2 + \dots + b^d$$
- No of nodes generated in an iterative deepening search:

$$N_{IDS} = (d+1)b^0 + db^1 + (d-1)b^2 + \dots + b^d$$
- For $b = 10, d = 5$
 - $N_{BFS} = 1+10 + 100 + 1,000 + 10,000 + 100,000 = 111,111$
 - $N_{IDS} = 6+50 + 400 + 3,000 + 20,000 + 100,000 = 123,456$
- In general IDS is the preferred uninformed search method when there is a large search space and the depth of the solution is not known

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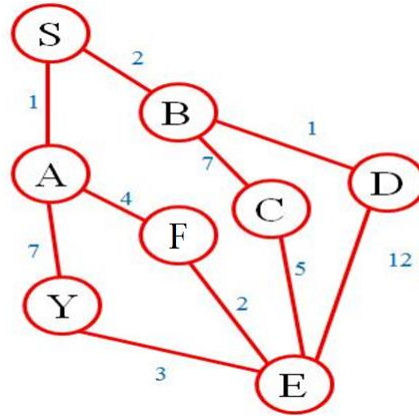
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Example

- Find path from S to E
- Using IDS



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Informed Search Strategies

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Informed (Heuristic) Searches

- One that uses problem specific knowledge beyond the definition of the problem itself
- Rather than trying all possible search paths, try to focus on paths that seems to be getting closer to goal using some kind of “guide”
- If not provided with a right guide, it may not give the correct solution

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Heuristic

- Greek word which means find or discover
- Refers to estimated or experience based problem solving approaches, learning and discovery
 - *e.g., a physician, a car mechanic*

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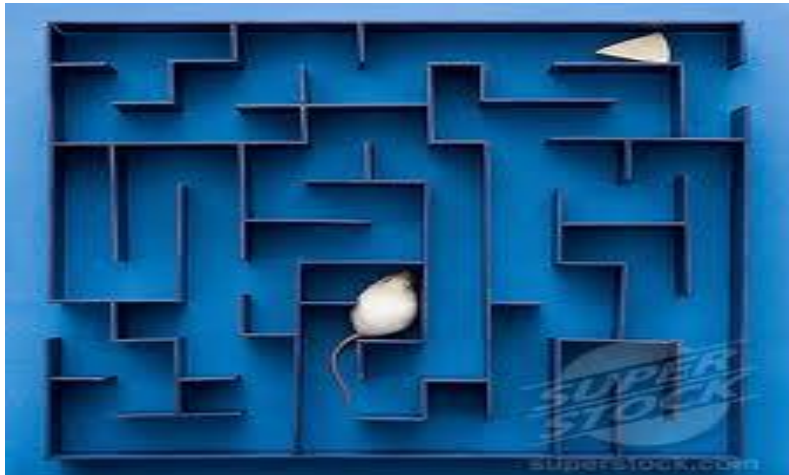
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Mouse in a Maze



The strength of smell of cheese guides the mouse. The smell of cheese is heuristic for mouse.

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Heuristic function

- Additional knowledge of the problem is imparted to the search algorithm.
- $h(n)$ = estimated cost of the cheapest path from node n to goal
- If n is the goal node then $h(n) = 0$.
- Value of this function guide us whether we are close to our goal

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Best First Search

- The node selected for expansion based on an evaluation function $f(n)$.
- The node with the lowest evaluation (cost) is selected for expansion
- It can be implemented using a priority queue
- The fringe shall contain the f values in ascending order
- Special cases
 - Greedy Best First Search
 - A* Search

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Greedy Best First Search

- Tries to expand a node which is closest to the goal
- $f(n) = h(n)$
- $f(n)$ = Estimated cost from n to goal
- $h_{SLD}(n)$ = the straight line distance from n to goal (Bucharest)
- $h_{SLD}(n)$ is an underestimate of the actual distance

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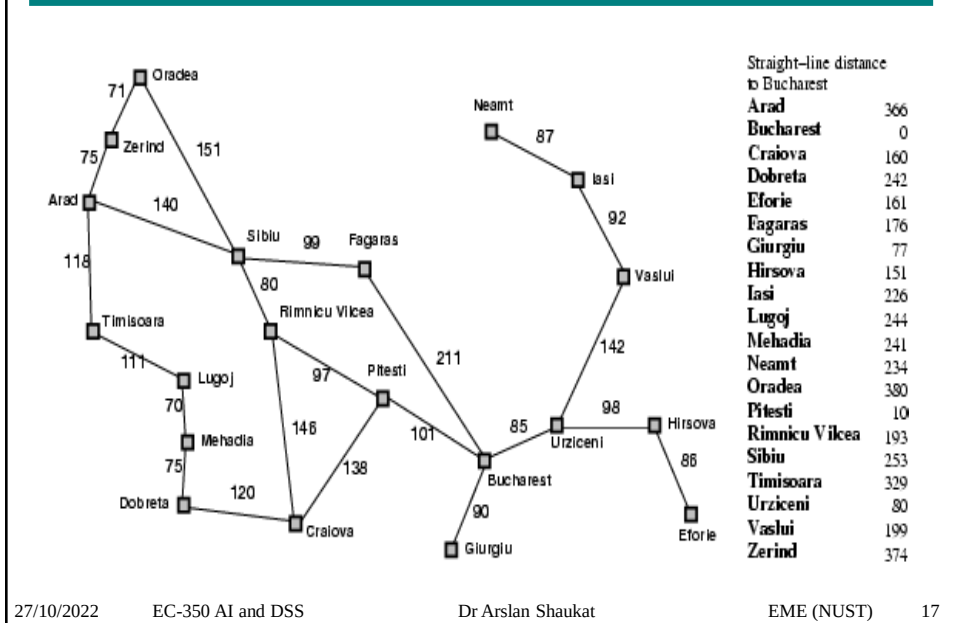
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Romania with Step Costs in km



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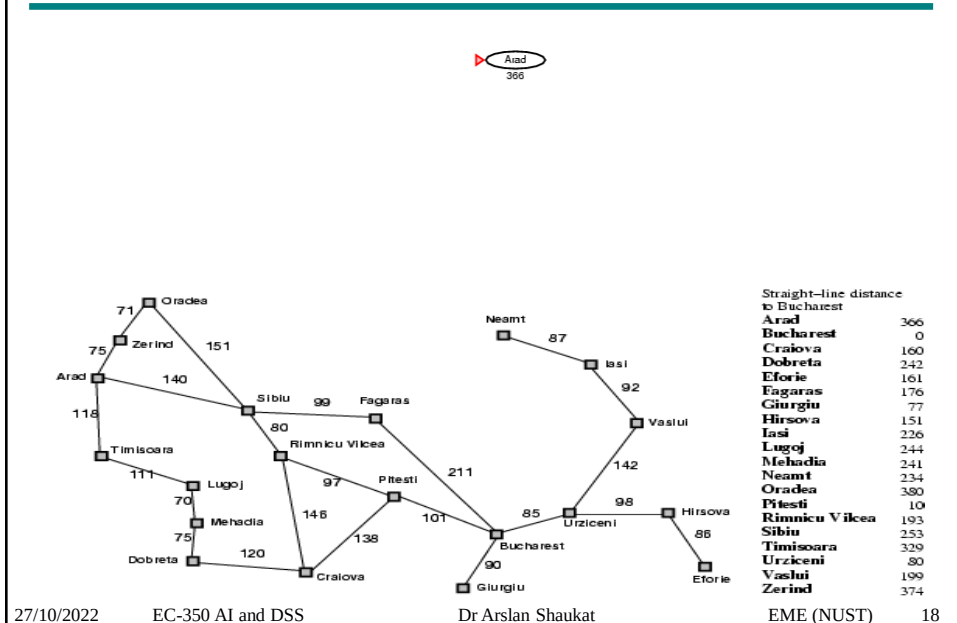
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Greedy best-first search example



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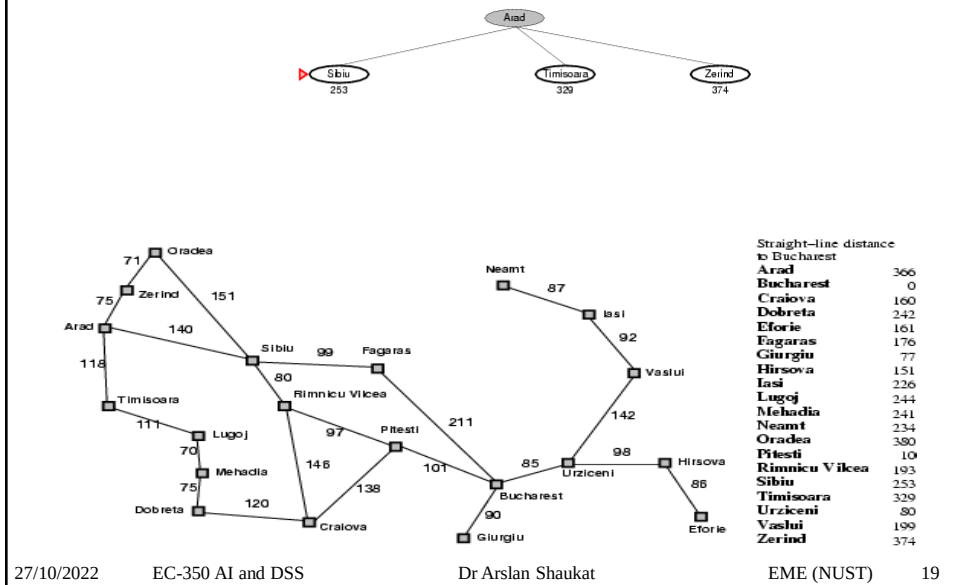
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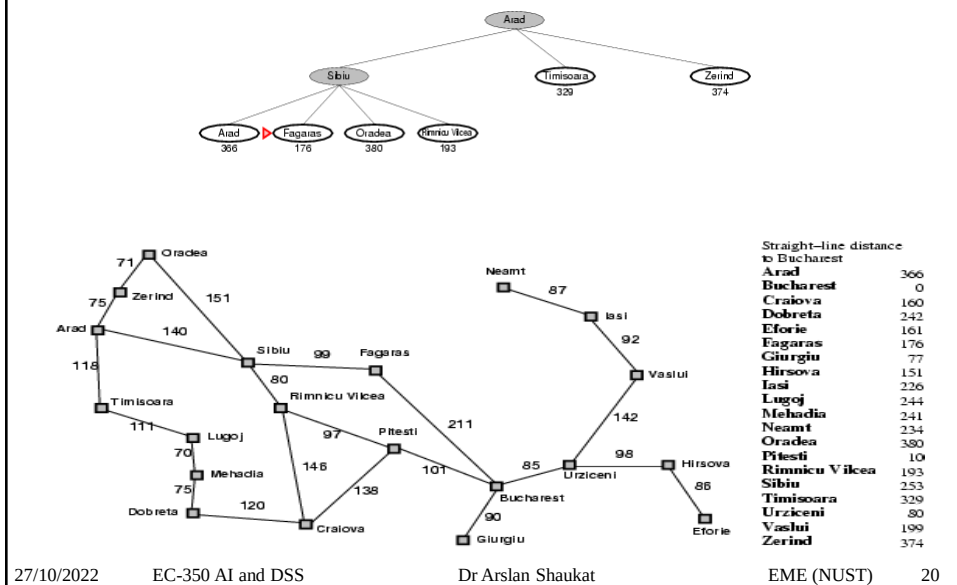
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Greedy best-first search example



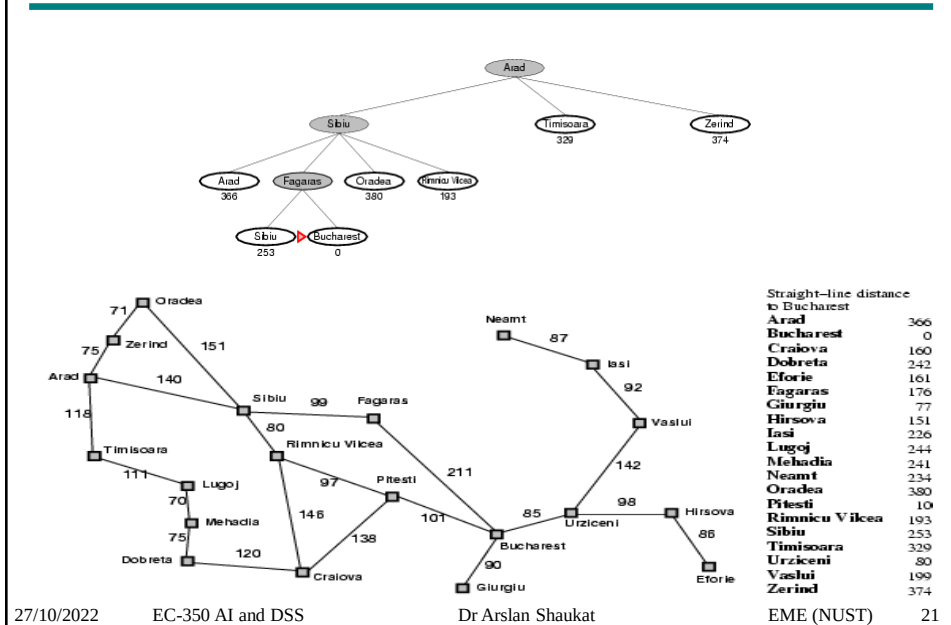
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Greedy best-first search example



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Greedy best-first search example



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Properties of Greedy Best-First Search

- Resembles the depth-first search in a way that it prefers to follow a single path all the way to the goal
- Complete?** No – can get stuck in loops, e.g., Iasi → Neamt → Iasi → Neamt →
- Time?** $O(b^m)$, but a good heuristic can give dramatic improvement
- Space?** $O(b^m)$ -- keeps all nodes in memory
- Optimal?** No

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A* Search

- It evaluate nodes by introducing another function $g(n)$
- $g(n)$ = cost from start node to reach n
- $h(n)$ = estimated cost from n to goal
- $f(n)$ = estimated total cost of path through n to goal
- Evaluation function $f(n) = g(n) + h(n)$

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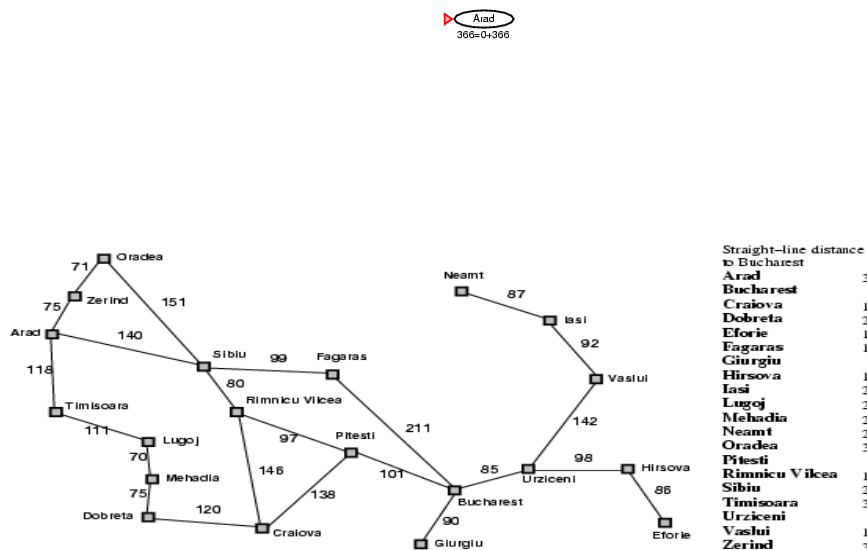
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A* search example



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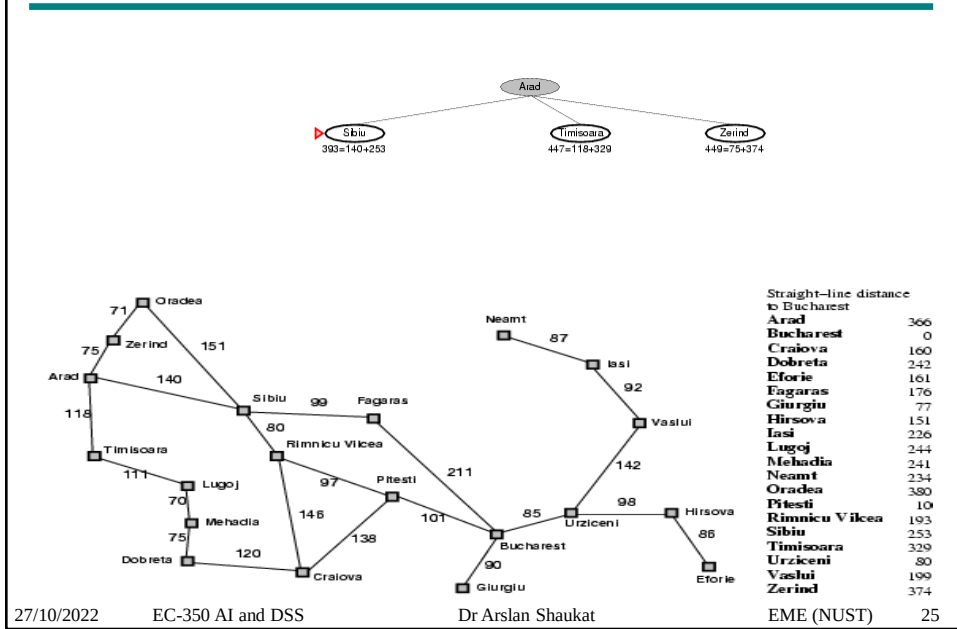
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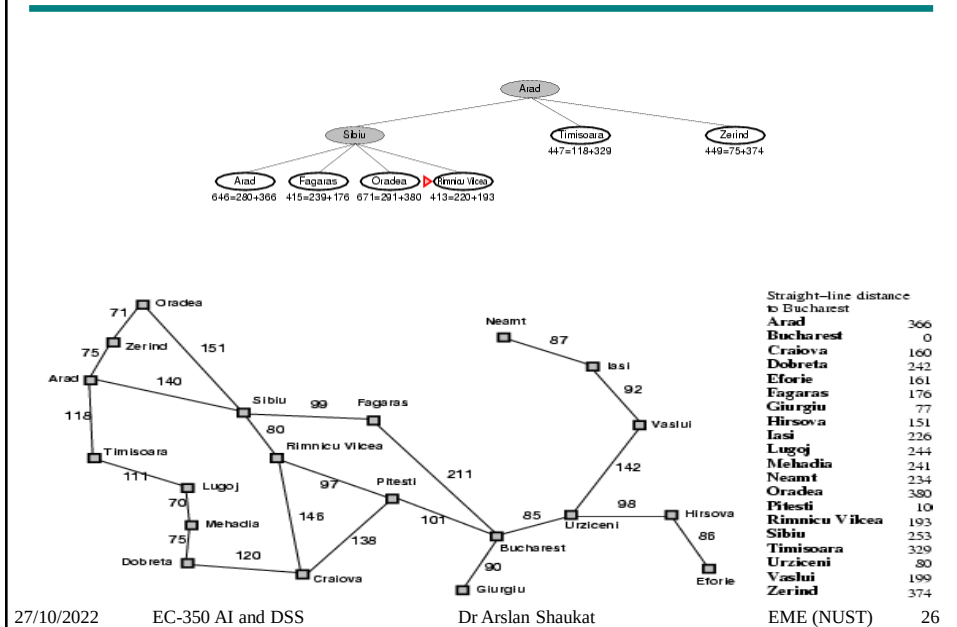
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A* search example



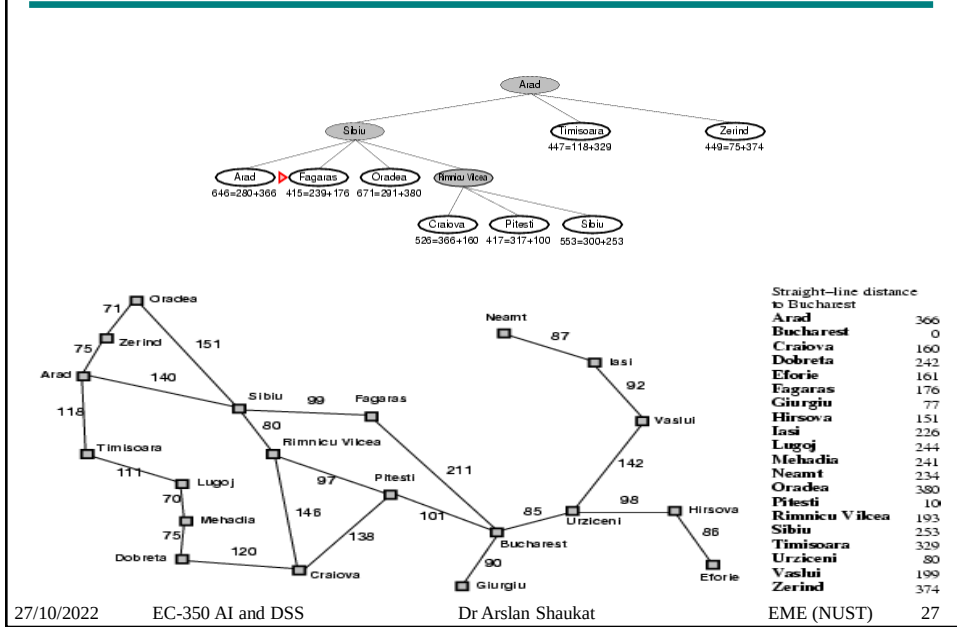
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A* search example



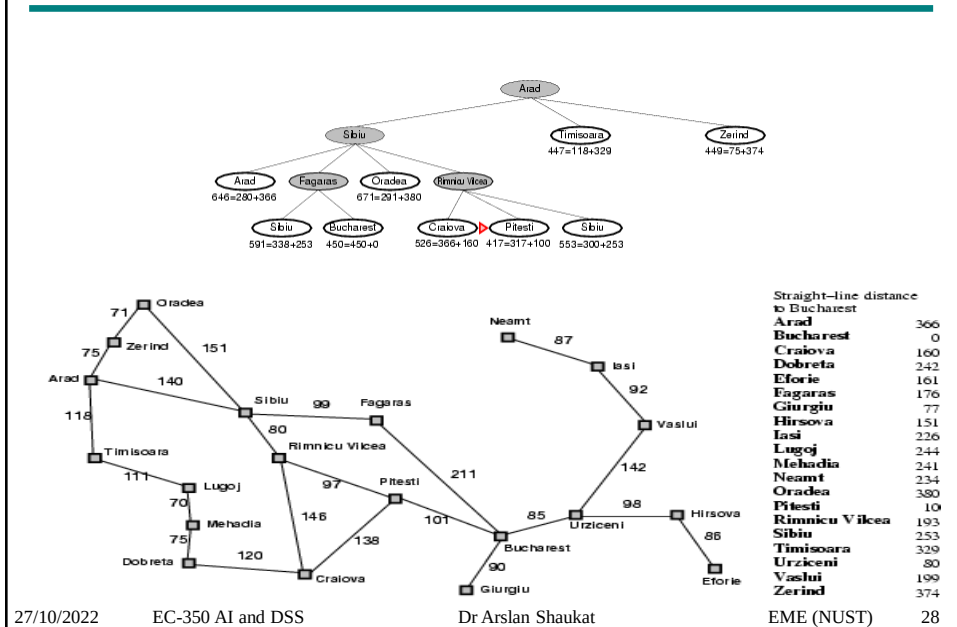
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A* search example



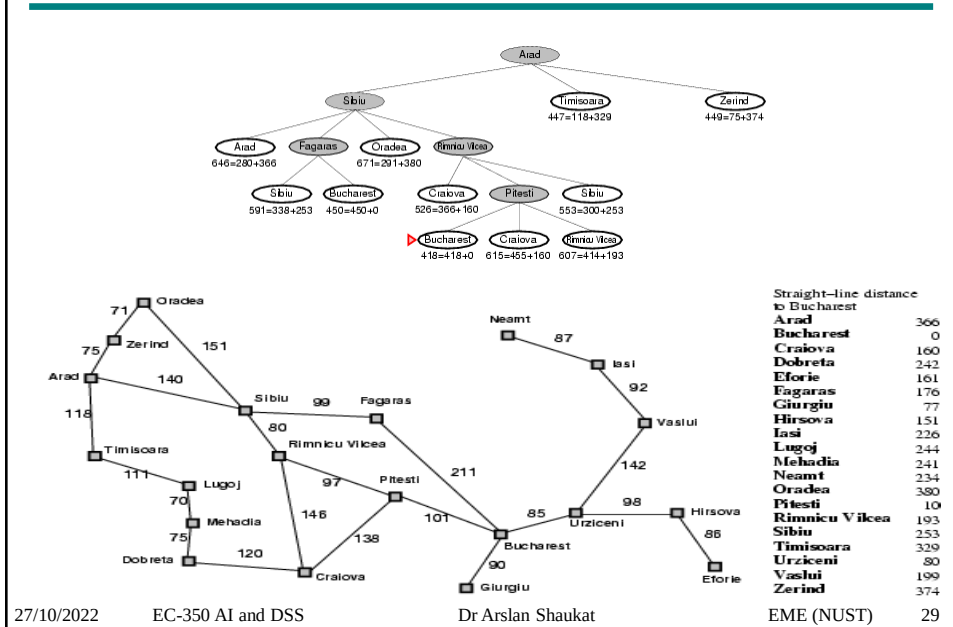
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A* search example



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A* search example



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Properties of A*

- **Complete?** Yes (unless there are infinitely many nodes)
- **Time?** Exponential in d
- **Space?** $O(b^m)$ Keeps all nodes in memory
- **Optimal?** Yes
 - **Note:** A* is also optimally efficient for a given heuristic: That is, no other optimal search algorithm is guaranteed to open fewer nodes

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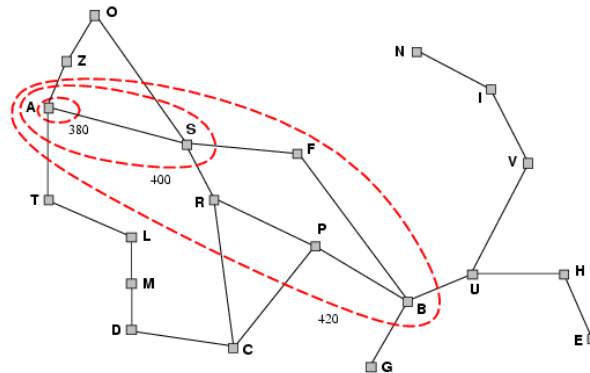
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Optimality of A*

- A* expands nodes in order of increasing f value
- Gradually adds " f -contours" of nodes



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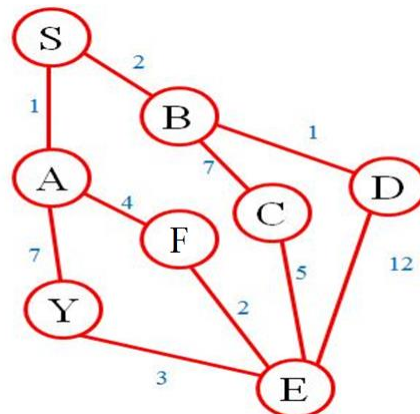
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Example

- Find the path from S to E
- Using Greedy best first search
- Use A* search



Values for h:

A:5, B:6, C:4, D:15, F:5, Y:8, E:0

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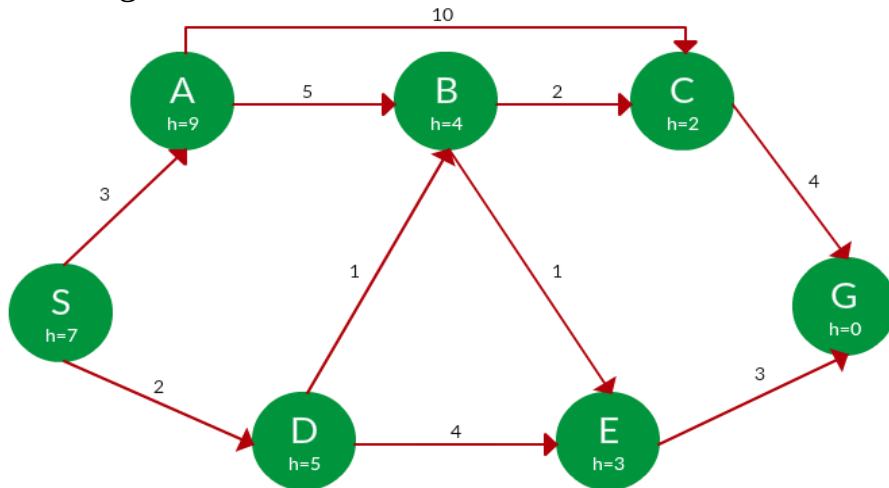
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Example

- Find the path from S to G
- Using GBFS and A* search



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